

[Academic Presentation Title]

[Presentation Subtitle]

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[Conference Name, Month Year]

Progressive vs. Non-progressive Visualization

Non-progressive visualization

- Renders in one pass.
- Each change in the data, whether caused by interaction or new data, triggers a full rendering.

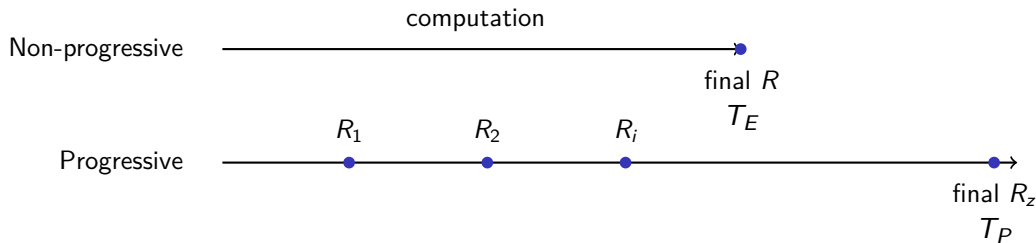
Progressive visualization

- Rendering uses a series of passes that improve the result until completion.
- Each pass is limited to a requested latency and renders a meaningful partial result.
- Each partial result has a meaningful error or quality measure and an estimate of completion, such as a progress bar.
- Users can interact at any time; the visualization adapts as much as possible and can implement steering when needed.

Definition adapted from Ulmer et al., “A Survey on Progressive Visualization” (2024).

Earlier Results Do Not Guarantee Faster Completion

Schematic possibility (not to scale): incremental-update overhead can make PVA finish later [4].



Partial results $R_1, R_2, \dots$ are available for analysis before the progressive run finishes.

Total runtime depends on the implementation.

Update overhead can extend completion [4]; A-tSNE reports computational gains [2].

Progressive Computation

F_p : progressive function

q : requested interval between outputs

R_{i-1}, R_i, R : previous, partial, and final result

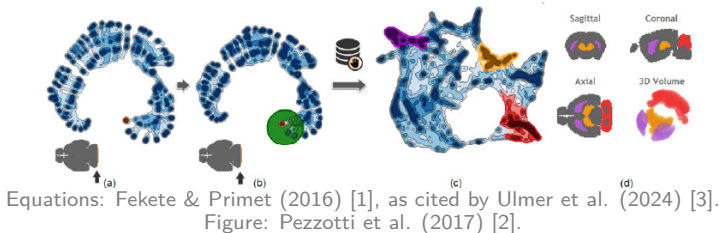
P : function parameters; D_i : data at pass i

t_i, m_i : time and memory at pass i

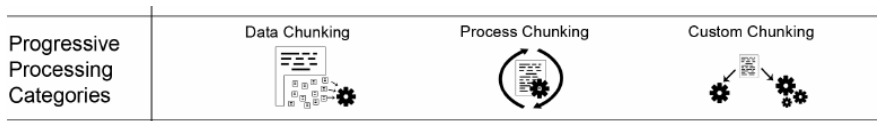
$R_0 = \emptyset$: first call has no previous result

$$F_p(P, D_i, q, R_{i-1}) \xrightarrow{t_i, m_i} R_i, \quad i = 1, \dots, z$$

$t_i \leq q$, $R_i \rightarrow R$ or a termination criterion is reached.



Three Ways to Produce Progressive Results



Ulmer et al. (2024), "A Survey on Progressive Visualization."

Data chunking

$$D_i = d_1 \cup \dots \cup d_i$$
$$R_i = F_p(P, D_i, q, R_{i-1})$$

R_i summarizes the data processed so far. It may be incomplete for the full dataset.

Process chunking

$$D_i = D$$
$$R_i = F_p(P, D, q, R_{i-1})$$

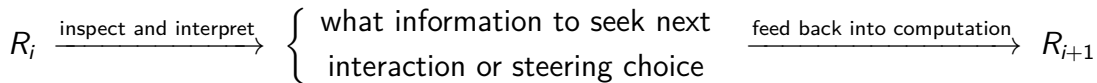
R_i is the current iteration output on the full dataset.

Custom chunking

$$D_i = \pi_i(P, D, R_{i-1})$$
$$R_i = F_p(P, D_i, q, R_{i-1})$$

R_i follows a task-specific or hybrid schedule.

From Intermediate State to the Next User Action



Analyzing R_i informs both what the user seeks next and how the process is steered.

Potential decision signals in the intermediate states

Quality / error Uncertainty Change Stability / convergence

Progressive Visual Analytics: Two Perspectives

Ulmer et al. (2024)

Latency-bounded passes. Each pass returns a meaningful partial result within a requested latency.

Quality feedback. Partial results carry quality or error information and an estimate of completion.

Ongoing interaction. Users can interact throughout the process; the visualization adapts and can support steering.

“A Survey on Progressive Visualization” [3].

Pezzotti et al. (2017)

Early intermediate results. Approximate initialization lets users begin interpreting the embedding before exact computation finishes.

User-steered refinement. Visual feedback on approximation supports local refinement and control of the approximation level.

Updates without restarting. Data can change during optimization while approximate neighborhoods are refined in the background.
“Approximated and User Steerable tSNE for Progressive Visual Analytics” [2].

Paper 1: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 1: Motivation

- *[Why the research question matters]*
- *[Limitations of existing explanations]*
- *[How the paper approaches the question]*

Paper 1: Research Questions and Hypotheses

- *[Research question]*
- *[Hypothesis or proposition 1]*
- *[Hypothesis or proposition 2, if applicable]*

Paper 1: Conceptual Framework

- *[Key concepts or variables]*
- *[Theoretical relationships and proposed mechanisms]*
- *[Insert the paper's central framework figure here]*

Paper 1: Data and Research Subjects

- *[Data or material sources and time period]*
- *[Sample composition or case selection]*
- *[Data processing and ethics, if applicable]*

Paper 1: Research Methods

- *[Research design and analytical method]*
- *[Model, experiment, or analysis workflow]*
- *[Key identification assumptions or derivation conditions]*

Paper 1: Results Overview

- *[Main result 1]*
- *[Main result 2]*
- *[How the results address the hypotheses or propositions]*

Paper 1: Main Evidence

- *[Insert the key figure, table, or theorem]*
- *[Key estimate, proof, or observation]*
- *[Conclusion supported by this evidence]*

Paper 1: Mechanisms and Additional Analyses

- *[Mechanism analysis or theoretical explanation]*
- *[Robustness, sensitivity, or alternative explanation checks]*
- *[Boundary conditions for the results]*

Paper 1: Summary

- *[Main conclusion]*
- *[Contribution and limitation]*
- *[Connection to Paper 2]*

Paper 2: Overview

- *[Title, authors, and publication details]*
- *[Research subject and context]*
- *[Core question addressed in this paper]*

Paper 2: Research Question and Contribution

- *[Research question]*
- *[What this paper adds to related work]*
- *[Relationship to or distinction from Paper 1]*

Paper 2: Framework and Hypotheses

- *[Theoretical basis or conceptual framework]*
- *[Hypotheses or propositions]*
- *[Proposed mechanism or reasoning]*

Paper 2: Data and Research Design

- *[Data or material sources and sample]*
- *[Research design and key variables or concepts]*
- *[Why the design fits the research question]*

Paper 2: Methods and Analysis Strategy

- *[Model, experiment, comparison, or proof method]*
- *[Identification strategy or analytical steps]*
- *[Key assumptions and how they are assessed]*

Paper 2: Main Results

- *[Main result 1 and supporting evidence]*
- *[Main result 2 and supporting evidence]*
- *[How the results answer the research question]*

Paper 2: Mechanisms and Robustness

- *[Mechanisms, heterogeneity, or boundary conditions]*
- *[Robustness or sensitivity analyses]*
- *[Alternative explanations and additional evidence]*

Paper 2: Summary and Overall Conclusions

- *[Main conclusion of Paper 2]*
- *[Shared findings and differences across the two papers]*
- *[Overall answer to the research topic]*

- [1] J.-D. Fekete and R. Primet, “Progressive Analytics: A Computation Paradigm for Exploratory Data Analysis,” CoRR, abs/1607.05162, 2016.
- [2] N. Pezzotti et al., Approximated and User Steerable t-SNE for Progressive Visual Analytics, *IEEE Transactions on Visualization and Computer Graphics*, 23(7), 1739–1752, 2017. doi:10.1109/TVCG.2016.2570755.
- [3] A. Ulmer et al., A Survey on Progressive Visualization, *IEEE Transactions on Visualization and Computer Graphics*, 30(9), 6447–6467, 2024. doi:10.1109/TVCG.2023.3346641.
- [4] M. Angelini, G. Santucci, H. Schumann, and H.-J. Schulz, “A Review and Characterization of Progressive Visual Analytics,” *Informatics*, 5(3), 31, 2018. doi:10.3390/informatics5030031.

[Thank You]

[Questions and Discussion]

[Presenter email, optional]